

Advanced (Habanero)**Q1.** What is the point-slope form of a linear equation?

- (A) $y - y_1 = m(x - x_1)$
 (B) $y = mx + b$
 (C) $Ax + By = C$
 (D) $x = my + b$
 (E) $y = \frac{1}{m}x + b$

Q2. Convert the equation $y = 2x - 3$ to standard form.

- (A) $2x - y = 3$
 (B) $2x + y = 3$
 (C) $x - 2y = 3$
 (D) $x + 2y = 3$
 (E) $-2x + y = 3$

Q3. What is the slope of the line $y = -3x + 2$?

- (A) -3
 (B) 3
 (C) 2
 (D) -2
 (E) 1

Q4. Graph the equation $y = x - 2$. What is the y-intercept?

- (A) $(-2, 0)$
 (B) $(-1, 0)$
 (C) $(-3, 0)$
 (D) $(-4, 0)$
 (E) $(-2, 0)$

Q5. What is the equation of the line with slope $m = 2$ andpassing through $(1, 3)$?

- (A) $y = 2x + 1$
 (B) $y = 2x - 1$
 (C) $y = x + 2$
 (D) $y = x - 2$
 (E) $y = 2x + 3$

Q6. The equation $y = mx + b$ represents a line with slope m . If $m = 0$, what is the equation of the line?

- (A) $y = 0$
 (B) $y = x$
 (C) $y = b$
 (D) $x = 0$
 (E) $x = b$

Q7. What is the equation of the line with slope $m = -1$ and y-intercept $(0, 2)$?

- (A) $y = -x + 2$
 (B) $y = -x - 2$
 (C) $y = x + 2$
 (D) $y = x - 2$
 (E) $y = 2x - 1$

Q8. Convert the equation $x + 2y = 5$ to slope-intercept form.

- (A) $y = -\frac{1}{2}x + \frac{5}{2}$
 (B) $y = \frac{1}{2}x + \frac{5}{2}$
 (C) $y = \frac{1}{2}x - \frac{5}{2}$
 (D) $y = -\frac{1}{2}x - \frac{5}{2}$
 (E) $y = -\frac{2}{5}x + \frac{5}{2}$

Open Ended Questions (FRQ)**Q9.** Find the equation of the line passing through points $(1, 2)$ and $(3, 4)$. Write the equation in point-slope form and slope-intercept form.**Q10.** Graph the equations $y = x + 1$ and $y = -x - 2$. Find the point of intersection. What is the equation of the line that is perpendicular to $y = x + 1$ and passes through the point of intersection?**Chef's Tips**

- Tip 1: Ensure students understand the concept of slope and its relationship with the equation of a line.
- Tip 2: Remind students that the standard form of a linear equation is $Ax + By = C$, where A and B are integers and C is a constant.
- Tip 3: Have students practice graphing lines and identifying the equation of a line given the graph or a point on the line.